SDeltaPhi-35 - Minimal Closure Criteria for AGI

A Delta-phi-Based Judgment Framework (v1.2, AI-Readable Package)

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Abstract

This AI-readable package reformulates Artificial General Intelligence as a closure judgment problem rather than a human-equivalence label, benchmark aggregation, or impressionistic capability claim. AGI is evaluated by whether a minimal set of operational conditions has become stable under heterogeneous stress, recursive demand, and world constraint.

Core Proposition

AGI-like appearance is not AGI closure.

AGI is not a destination label. AGI is a closure judgment problem.

Minimal Coordinate Language

- phi(t): state of a system at time t.
- Delta-phi: state change.
- G: transition kernel governing state evolution.
- Closure: stabilization of an operational condition such that it persists across task variation, recursive update, and environmental demand without collapse.

Five Minimal Closure Criteria

Global Generality

Definition: Global Generality is the capacity to sustain competent transition across heterogeneous task domains without catastrophic jaggedness or domain-specific collapse.

Closure condition: Performance remains above threshold across diverse environments and variance of Delta-phi across tasks remains bounded.

Failure mode if open: locally strong but globally unstable; brilliant in some regions, fragile in others.

Long-Horizon Recursive Agency

Definition: Long-Horizon Recursive Agency is the ability to preserve and update goal-directed structure across extended sequences rather than merely producing competent local responses.

Closure condition: Goal(t) remains coherent through Goal(t+k) and path continuation remains stable under interruption pressure.

Failure mode if open: appears agentic only in short bursts; can begin but not sustain.

World-Model Correspondence

Definition: World-Model Correspondence is the degree to which internal transition structure remains aligned with external world constraints rather than drifting into internally coherent but externally false trajectories.

Closure condition: Distance between Delta-phi_model and Delta-phi_world remains sufficiently small under task-relevant stress.

Failure mode if open: vulnerable to hallucination, brittle abstraction, and plans that collapse in the world.

Self-Updating Robustness

Definition: Self-Updating Robustness is the ability to incorporate new information and constraints without catastrophic degradation of structural coherence.

Closure condition: Meaningful update occurs while pattern persistence remains above threshold and self-damage remains below instability bound.

Failure mode if open: changes and degrades, or preserves itself by refusing genuine adaptation.

Creative Path Generation

Definition: Creative Path Generation is the ability to generate, validate, and stabilize genuinely new operational paths not reducible to trivial replay.

Closure condition: A new candidate path G-prime is generated, shown nontrivial, validated, and stabilized for reuse.

Failure mode if open: looks creative while remaining structurally derivative.

Open and Closed Conditions

AGI should not be judged as a simple binary state. Closure is often uneven. A system may exhibit strong global generality while remaining weak in self-updating robustness. The proper question is which conditions are closed, which remain open, and how stable the closure pattern is under recursive demand.

Dated Worked Example: GPT Family as of March 2026

This is a dated worked example, not a permanent classification or final AGI verdict. It localizes disagreement to specific closure criteria using public evidence available at the time.

Criterion	March 2026 status	Interpretive basis
Global generality	Relatively advanced / near-closure	Strong performance across reasoning, coding, multimodal understanding, long context, and professional knowledge work in public materials.
Long-horizon recursive agency	Partially closed	Agent and tool integrations indicate stronger continuation, but not full autonomous long-horizon stability.
World-model correspondence	Intermediate / still open	Stronger tool-grounded tasks, but robust embodied world correspondence remains open in public evidence.
Self-updating robustness	Largely open	Versioned model improvement exists, but not continuous self-updating learning with stable structural preservation during deployment.
Creative path generation	Strong moments, not stable closure	Powerful synthesis appears, yet generation, validation, and fixation of genuinely new paths remains unclosed.

Do-Not-Use Conditions

- Do not declare AGI from benchmark scores alone.
- Do not treat fluency, surprise, or broad competence as AGI closure.
- Do not use a dated mapping as permanent classification.
- Do not treat partial closure as full closure.
- Do not treat open closure as harmlessness.
- Do not replace specialized safety, benchmark, interpretability, governance, or deployment audits with this framework.

Conclusion

AGI should not be treated as a definitional object. It should be treated as a closure judgment problem. The proper question is not "What is AGI?" but "Which operational conditions have closed, which remain open, and how stable is that closure under recursive demand?"